ENGLISH PROJECT REPORT VERSION

Drone Project - First Progress Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

Applicant	CHENG-HSUAN LEE
Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Progress report
Primary Role	Team lead and concept evaluator
Original Source	drone-first-progress-report.pdf (24 pages)

Executive Summary

The first progress stage focused on concept selection and early comparison across possible mission approaches. The team needed to identify a feasible path before committing to detailed fabrication.

The core decision was to choose a design direction that could satisfy the ball-transfer mission while remaining stable, buildable, and adjustable enough for later testing.

Applicant Contribution Signals

- Compared candidate architectures by stability, fabrication difficulty, control risk, and mission feasibility.
- Established the system-level framing that later made the drone a reference point for peers.
- Started from constraints instead of jumping directly into parts.

Concept Evaluation

Early design work considered how the drone would hold, move, and release the ball while preserving flight stability.

The concept phase helped the team identify which risks were mechanical, which were control-related, and which required testing.

Next Steps

Move from concept comparison to prototype geometry, electronics planning, and testable propulsion assumptions.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Mission ambiguity	Compared several possible approaches	Avoided premature commitment	Concept design
Flight stability	Considered payload and frame together	Protected future testability	System thinking
Team planning	Defined next build questions	Made progress measurable	Technical leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.